

Product Requirement Document (PRD)

Project Name: Danger Zone Mapping and Historical Information System

Version: 1.2

Prepared by: Gabor, Manidllangan, Pace

Date: Updated 11/26/2025

1. Introduction

1.1 Purpose

The purpose of the Danger Zone Mapping and Historical Information System is to provide an interactive desktop platform that identifies and visualizes areas vulnerable to natural and man-made hazards such as floods, fires, and landslides. Through a user-friendly JavaFX GUI, the system enables users to explore risk zones through interactive map-based coordinate selection, review historical incident data, and access safety information. It aims to enhance public awareness, support disaster preparedness, and assist local authorities in decision-making and planning through both traditional data management and innovative map-based interaction.

1.2 Scope

- Displaying interactive maps showing categorized danger zones with real-time visual overlays.
 - Providing dual data management approaches: text-based CRUD operations and map-based interactive marker placement.
 - Interactive coordinate selection through direct map clicking with real-time feedback.
 - Adjustable hazard zone radius visualization (100-2000 meters) with live circle rendering.
 - Viewing detailed historical information on past disasters or incidents with filtering capabilities.
 - Providing search and filter options for specific locations or hazard types.
 - Allowing administrators to update or add new hazard records through form-based entry or map interaction.
 - Offering users essential safety guidelines and references for disaster prevention.
-

2. Objectives

- Provide a visual representation of hazard-prone areas using an interactive JavaFX GUI map.
 - Enable precise location marking through direct map interaction with coordinate capture.
 - Visualize hazard zones with adjustable radius circles and severity-based color coding.
 - Support dual data entry: traditional forms and intuitive map-based coordinate selection.
 - Centralize historical disaster data for improved accessibility and analysis.
 - Help residents and authorities identify risk zones and understand past incident patterns.
 - Support disaster preparedness and planning through accessible and organized information.
 - Promote community safety awareness through educational hazard references.
-

3. Target Users

- Residents - to identify nearby hazard-prone areas and access safety information.
- Local Government Units (LGUs) - to analyze risk zones, manage hazard data, and plan disaster responses.
- Disaster Management Offices - to document historical incidents and use data for preparedness strategies.

- Researchers and Students – to study hazard patterns and their historical impact within the locality.
-

4. Functional Requirements

4.1 Input & Configuration

- Users can input or update hazard data such as type, location, and date of incident.
- Administrators can configure and manage map layers and hazard categories.
- Allows importing or exporting data from CSV or database sources.
- Supports manual entry of safety guidelines and references.
- Users can click on the map to capture precise latitude/longitude coordinates.
- Real-time coordinate display shows mouse position during map hover.
- Hazard radius adjustment through slider control (100-2000 meters) with live visual feedback.
- Selected coordinates persist with visual marker indicator.

4.2 Core Features

A. Danger Zone Mapping

- Display categorized hazard zones (e.g., flood, landslide, fire).
- Visualize map layers with legends and color-coded markers.
- Allow users to zoom, pan, and click for details.

B. Historical Information Viewer

- Show past incidents with date, location, type, and damage details.
- Support filtering by year, hazard type, or location.

C. Data Management (Admin Only)

- Add or update hazard information with description, coordinates, and severity.
- Click map to capture coordinates for new landmarks automatically.

D. Safety Guidelines Section

- Display preventive measures, contact numbers, and evacuation points.

E. Search and Filter

- Search by location name or hazard type.
- Filter map view based on user-selected criteria.

4.3 Output & Reporting

- Shows detailed information for each hazard area, including description and historical data.
 - Displays map overlays with hazard density and affected zones.
 - Generates summaries of past incidents by category or year.
 - Provides printable or exportable reports for disaster management use.
 - Optionally supports data visualization charts for hazard frequency or impact.
-

5. Non-Functional Requirements

- Usability: The interface must be intuitive and easy to navigate with immediate visual feedback.
- Performance: The system should load and display map data efficiently, even with multiple hazard layers.
- Reliability: Must ensure data accuracy and prevent data loss during updates or modifications.
- Scalability: Should support future expansion for additional regions or hazard categories.
- Security: Access control must restrict administrative functions to authorized users only.

- Maintainability: The codebase should be modular and well-documented for future updates.
 - Portability: Compatible with major desktop operating systems running Java (Windows, macOS, Linux).
-

6. System Architecture

6.1 High-Level Architecture

[Java GUI Interface] → [Interactive Map Component] → [Data Controller] → [Database] → [Map Visualization Module]

6.2 Module Descriptions

- JavaFX GUI Interface: Main user interface for map interaction and data viewing.
 - Interactive Map Component: Handles coordinate capture, marker rendering, and radius visualization.
 - Data Controller: Handles user inputs, database queries, and updates.
 - Database: Stores hazard information, historical records, and map data.
 - Map Visualization Module: Renders interactive maps and hazard layers.
-

7. Technology Stack

- Frontend (Interface): JavaFX
 - Backend: Java
 - Database: MySQL
 - Mapping Library: JavaFX shapes with coordinate conversion system
 - Hosting: Local execution on desktop
-

8. Workflow

- User launches the Java application.
 - System loads local database and map interface.
 - User searches or clicks map to select hazard type/location.
 - Map displays relevant danger zones and historical data with color-coded markers.
 - User can click map to capture coordinates for new entries.
 - User adjusts radius slider to define hazard zone size with live preview.
 - Admin can log in to update or add new hazard records via forms or map interaction.
 - Reports or data summaries can be viewed or exported.
-

9. Deliverables

- Executable Java GUI Application (.jar)
 - User Manual (for residents and LGU users)
 - Administrator Guide (for managing hazard data)
 - Database Schema and setup instructions
 - Sample Data Files (for demo and testing)
 - Ormoc City Map Image (PNG, 1920×988 pixels)
-

10. Roadmap

Phase 1: Core Development

- GUI design and basic map visualization
- Database integration and data display

Phase 2: Enhancements

- Admin panel for data management
- Search and category filters
- Interactive map coordinate capture

Phase 3: Integration

- Report generation and data export
 - API integration for live updates
-

11. Risks & Constraints

- Accuracy depends on input data from local authorities.
 - Map visualization limited to static image with coordinate conversion.
 - Does not provide real-time disaster alerts or predictions.
 - Requires regular manual updates for latest hazard data.
-

12. Success Criteria

- Displays categorized danger zones accurately on the map.
- Users can add landmarks by clicking map with precise coordinate accuracy.
- Hazard zones display with adjustable radius (100-2000m) and real-time visual feedback.
- Loads historical disaster data reliably.
- Enables admin users to manage hazard information via text forms and map interaction.